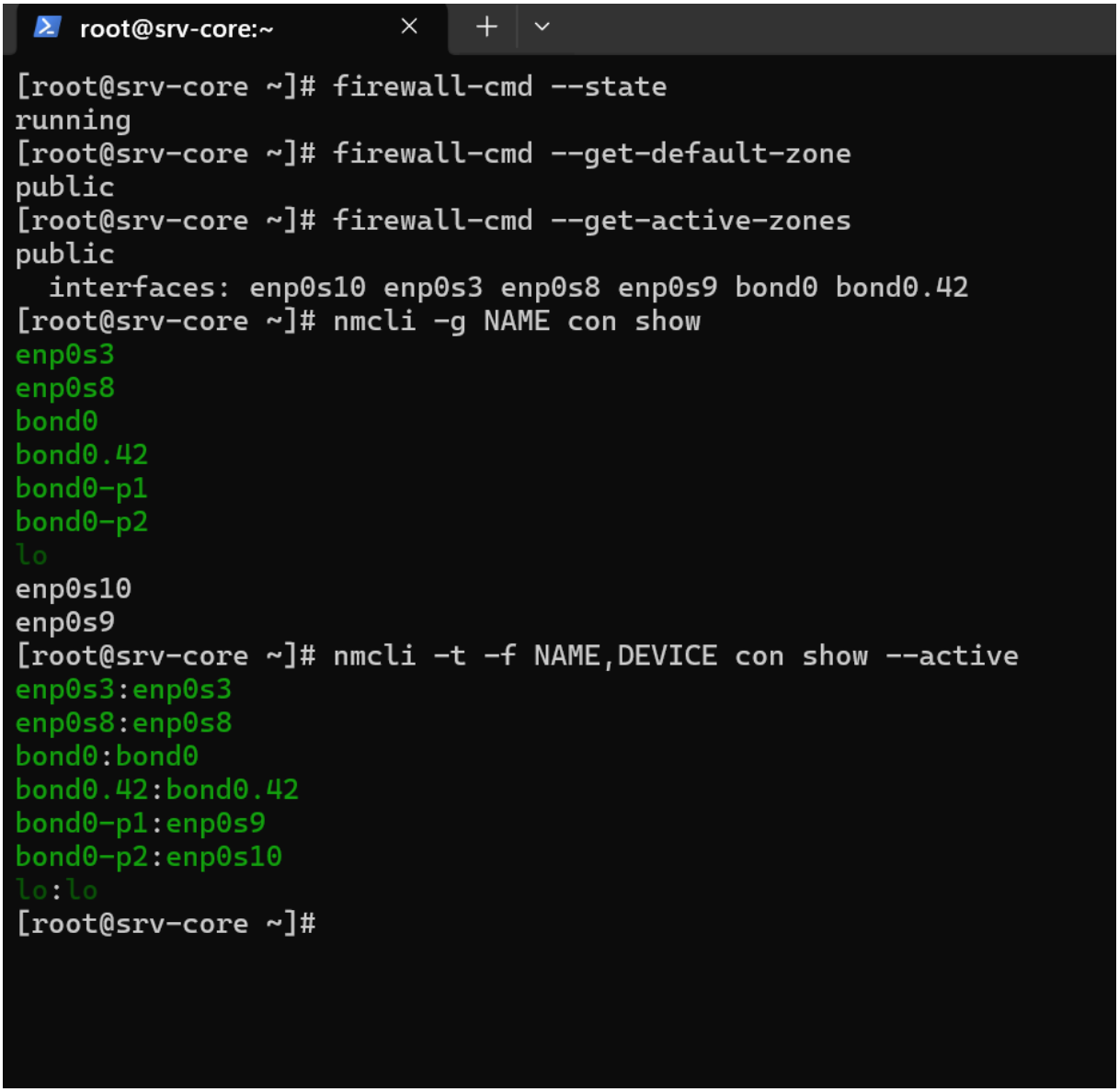
TP3 — Sécurisation d'un serveur Linux

Mission 1 — Pare-feu firewalld segmenté par zone :

1.1 Remise en service et état des lieux :

Cette capture présente l'état initial du pare-feu ainsi que les interfaces réseau actives. Elle permet de vérifier la configuration avant les modifications.



Mission 1.2 :

Affectation de chaque interface à sa zone :

Les interfaces réseau sont associées aux zones Firewalld appropriées. Chaque réseau bénéficie ainsi de règles de sécurité adaptées à son usage.

```
[root@srv-core ~]# nmcli con up enp0s3
Connexion activée (chemin D-Bus actif : /org/freedesktop/NetworkManager/ActiveConnectio
n/9)
[root@srv-core ~]# nmcli con up enp0s8
Connexion activée (chemin D-Bus actif : /org/freedesktop/NetworkManager/ActiveConnectio
n/10)
[root@srv-core ~]# nmcli con up bond0
Connexion activée (controller waiting for ports) (Chemin D-Bus actif : /org/freedesktop
/NetworkManager/ActiveConnection/11)
[root@srv-core ~]# nmcli con up bond0.42
Connexion activée (chemin D-Bus actif : /org/freedesktop/NetworkManager/ActiveConnectio
n/14)
[root@srv-core ~]# firewall-cmd --set-default-zone=drop
success
[root@srv-core ~]# firewall-cmd --get-active-zones
drop
  interfaces: enp0s3 enp0s10
internal
  interfaces: bond0 bond0.42
public
  interfaces: enp0s9
work
  interfaces: enp0s8
[root@srv-core ~]# nmcli -g NAME con show | while read -r c ; do
    printf '%s : %s\n' "$c" "$(nmcli -g connection.zone con show "$c")"
done
enp0s3 : drop
enp0s8 : work
bond0 : internal
bond0.42 : internal
bond0-p1 :
bond0-p2 :
lo :
enp0s10 :
enp0s9 :
[root@srv-core ~]#
```

Mission 1.4 :

Blacklist :

L'adresse IP est ajoutée à une liste noire dynamique. Les règles du pare-feu bloquent alors automatiquement les connexions provenant de cette source.

```
[root@srv-core ~]# firewall-cmd --get-ipsets
helios-blacklist
[root@srv-core ~]# firewall-cmd --zone=work --list-rich-rules
rule port port="10051" protocol="tcp" reject
rule service name="ssh" log prefix="SSH-NEW " level="info" limit value="3/m" accept limit
lue="6/m"
rule source ipset="helios-blacklist" drop
[root@srv-core ~]# firewall-cmd --check-config && firewall-cmd --reload
success
success
[root@srv-core ~]#
```

Mission 1.5:

La chaîne NFTables affichée permet de vérifier que les règles générées par Firewalld sont effectivement appliquées au niveau du noyau Linux.

```
[root@srv-core ~]# nft list tables
table inet firewalld
[root@srv-core ~]# nft list chain inet firewalld filter_IN_work
table inet firewalld {
        chain filter_IN_work {
                jump filter_INPUT_POLICIES_pre
                jump filter_IN_work_pre
                jump filter_IN_work_log
                jump filter_IN_work_deny
                jump filter_IN_work_allow
                jump filter_IN_work_post
                jump filter_INPUT_POLICIES_post
                meta l4proto { icmp, ipv6-icmp } accept
                meta pkttype host log prefix "filter_IN_work_REJECT: "
                reject with icmpx admin-prohibited
        }
}
```

Scan Nmap SRV-EDGE :

Le scan confirme que seuls les services explicitement autorisés restent accessibles depuis le réseau.

```
root@srv-edge:~
[root@srv-edge ~]# nmap -Pn -p 22,80,10051 192.168.56.10
Starting Nmap 7.92 ( https://nmap.org ) at 2026-11-04 14:21 CET
Nmap scan report for 192.168.56.10
Host is up (0.0036s latency).

PORT      STATE    SERVICE
22/tcp    open     ssh
80/tcp    filtered http
10051/tcp filtered zabbix-trapper
MAC Address: 08:00:27:2F:2E:56 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.31 seconds
[root@srv-edge ~]#
```

Mission 2 — Durcissement de sshd :

2.2 Fichier de surcharge :

Drop-in SSH :

Le fichier de surcharge permet de personnaliser la configuration SSH sans modifier le fichier principal distribué par le système.

```
[root@srv-core ~]# cat /etc/ssh/sshd_config.d/10-helios-hardening.conf

PubkeyAuthentication yes
PasswordAuthentication no
KbdInteractiveAuthentication no
PermitEmptyPasswords no
AuthenticationMethods publickey

MaxAuthTries 3
MaxSessions 4
MaxStartups 10:30:60
LoginGraceTime 30

AllowGroups ssh-admins

X11Forwarding no
AllowAgentForwarding no
AllowTcpForwarding no
AllowStreamLocalForwarding no
PermitTunnel no
GatewayPorts no
PermitUserEnvironment no

ClientAliveInterval 300
ClientAliveCountMax 2
TCPKeepAlive no

LogLevel VERBOSE
PrintLastLog yes

Banner /etc/issue.net

Subsystem sftp internal-sftp -l INFO -f AUTHPRIV

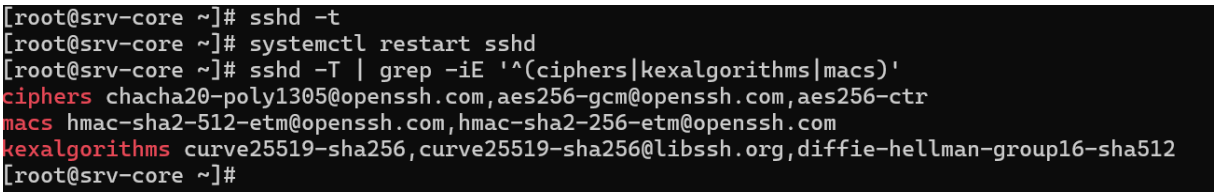
# Exception maitrisee : l'equipe exploitation a besoin d'un tunnel vers le port 10051
Match Group helios-ops
    AllowTcpForwarding yes
    PermitOpen localhost:10051
[root@srv-core ~]# sshd -t && echo "syntaxe OK"
syntaxe OK
[root@srv-core ~]#
```

2.3 Algorithmes : le piège crypto-policies

Validation SSH :

La commande sshd -t confirme l'absence d'erreur dans la configuration avant le redémarrage du service.





2.4 Changement de port : SELinux et pare-feu :

Changement de ports + Journal sshd :

Le service écoute désormais sur le nouveau port configuré. Les journaux confirment le bon démarrage après modification.

```
[root@srv-core ~]# semanage port -a -t ssh_port_t -p tcp 2242
[root@srv-core ~]# semanage port -l | grep '^ssh_port_t'
ssh_port_t                     tcp      2242, 22
[root@srv-core ~]# firewall-cmd --permanent --zone=work \
--add-rich-rule='rule port port="2242" protocol="tcp" log prefix="SSH-NEW " level="info" limit value="3/m" accept limit value="6/m"'
success
[root@srv-core ~]# firewall-cmd --permanent --zone=work \
--remove-rich-rule='rule service name="ssh" log prefix="SSH-NEW " level="info" limit value="3/m" accept limit value="6/m"'
success
[root@srv-core ~]# firewall-cmd --permanent --zone=internal --add-port=2242/tcp
success
[root@srv-core ~]# firewall-cmd --permanent --zone=internal --remove-service=ssh
success
[root@srv-core ~]# firewall-cmd --reload
success
[root@srv-core ~]# sshd -t
[root@srv-core ~]# systemctl restart sshd
[root@srv-core ~]# ss -ltnp | grep sshd
LISTEN 0      128          0.0.0.0:22        0.0.0.0:*    users:(("sshd",pid=7587,fd=7))
LISTEN 0      128             [::]:22           [::]:*    users:(("sshd",pid=7587,fd=8))
[root@srv-core ~]# journalctl -u sshd -n 20 --no-pager
nov. 04 14:46:24 srv-core systemd[1]: Starting OpenSSH server daemon...
nov. 04 14:46:24 srv-core sshd[6726]: Server listening on 0.0.0.0 port 22.
nov. 04 14:46:24 srv-core sshd[6726]: Server listening on :: port 22.
nov. 04 14:46:24 srv-core systemd[1]: Started OpenSSH server daemon.
nov. 04 14:49:14 srv-core sshd[6726]: Received signal 15; terminating.
nov. 04 14:49:14 srv-core systemd[1]: Stopping OpenSSH server daemon...
nov. 04 14:49:14 srv-core systemd[1]: sshd.service: Deactivated successfully.
nov. 04 14:49:14 srv-core systemd[1]: Stopped OpenSSH server daemon.
nov. 04 14:49:14 srv-core systemd[1]: Starting OpenSSH server daemon...
nov. 04 14:49:14 srv-core sshd[6868]: Server listening on 0.0.0.0 port 22.
nov. 04 14:49:14 srv-core sshd[6868]: Server listening on :: port 22.
nov. 04 14:49:14 srv-core systemd[1]: Started OpenSSH server daemon.
nov. 04 14:56:29 srv-core systemd[1]: Stopping OpenSSH server daemon...
nov. 04 14:56:29 srv-core sshd[6868]: Received signal 15; terminating.
nov. 04 14:56:29 srv-core systemd[1]: sshd.service: Deactivated successfully.
nov. 04 14:56:29 srv-core systemd[1]: Stopped OpenSSH server daemon.
nov. 04 14:56:29 srv-core systemd[1]: Starting OpenSSH server daemon...
nov. 04 14:56:29 srv-core sshd[7587]: Server listening on 0.0.0.0 port 22.
nov. 04 14:56:29 srv-core sshd[7587]: Server listening on :: port 22.
nov. 04 14:56:29 srv-core systemd[1]: Started OpenSSH server daemon.
[root@srv-core ~]#
```

2.5 Valider depuis srv-edge — sept tests :

Les sept tests SSH :

Cette série de tests vérifie les différents scénarios autorisés et interdits afin de valider la politique de sécurité définie.

```
sysops@srv-edge:~
[sysops@srv-edge ~]$ nc -zv -w2 192.168.56.10 22 ; echo "1. ancien port : echec attendu ($?)"
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: No route to host.
1. ancien port : echec attendu (1)
[sysops@srv-edge ~]$ /
-bash: / : est un dossier
[sysops@srv-edge ~]$ ssh -p 2242 sysops@192.168.56.10 'hostname; id -nG'
*** Serveur Helios Logistique - acces strictement reserve ***
Toute connexion est journalisee.
srv-core
sysops ssh-admins
[sysops@srv-edge ~]$ /
-bash: / : est un dossier
[sysops@srv-edge ~]$ ssh -p 2242 root@192.168.56.10 'id' ; echo "3. root : echec attendu ($?)"
*** Serveur Helios Logistique - acces strictement reserve ***
Toute connexion est journalisee.
root@192.168.56.10: Permission denied (publickey).
3. root : echec attendu (255)
[sysops@srv-edge ~]$ /
-bash: / : est un dossier
[sysops@srv-edge ~]$ ssh -p 2242 -o PubkeyAuthentication=no -o PreferredAuthentications=password \
 sysops@192.168.56.10 ; echo "4. mot de passe : echec attendu ($?)"
*** Serveur Helios Logistique - acces strictement reserve ***
Toute connexion est journalisee.
sysops@192.168.56.10: Permission denied (publickey).
4. mot de passe : echec attendu (255)
[sysops@srv-edge ~]$ /
-bash: / : est un dossier
[sysops@srv-edge ~]$ ssh -p 2242 nadia@192.168.56.10 ; echo "5. hors ssh-admins : echec attendu ($?)"
*** Serveur Helios Logistique - acces strictement reserve ***
Toute connexion est journalisee.
nadia@192.168.56.10: Permission denied (publickey).
5. hors ssh-admins : echec attendu (255)
[sysops@srv-edge ~]$ /
-bash: / : est un dossier
[sysops@srv-edge ~]$ ssh -p 2242 -N -L 9999:localhost:10051 sysops@192.168.56.10 & sleep 2
[1] 2560
*** Serveur Helios Logistique - acces strictement reserve ***
Toute connexion est journalisee.
[sysops@srv-edge ~]$ nc -zv -w2 127.0.0.1 9999 ; sleep 1 ; kill %1 2>/dev/null
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: Connected to 127.0.0.1:9999.
Ncat: 0 bytes sent, 0 bytes received in 0.02 seconds.
channel 2: open failed: administratively prohibited: open failed
[sysops@srv-edge ~]$ sleep 15
[1]+  Fini                    ssh -p 2242 -N -L 9999:localhost:10051 sysops@192.168.56.10
[sysops@srv-edge ~]$ ssh -p 2242 -o KexAlgorithms=diffie-hellman-group14-sha1 sysops@192.168.56.10
Unable to negotiate with 192.168.56.10 port 2242: no matching key exchange method found. Their offer: curve25519-sha256,curve25519-sha256@libssh.org,diffie-hellman-group16-sha512,
ext-info-s,kex-strict-s-v00@openssh.com
[sysops@srv-edge ~]$ echo "7. KEX SHA-1 : echec attendu ($?)"
7. KEX SHA-1 : echec attendu (255)
[sysops@srv-edge ~]$
```

SSHD -T :

```
[root@srv-core ~]# sshd -T | grep -iE 'permitrootlogin|passwordauthentication|allowgroups|ma
xauthtries|^port |loglevel'
port 2242
maxauthtries 3
permitrootlogin no
passwordauthentication no
loglevel VERBOSE
allowgroups ssh-admins
[root@srv-core ~]#
```

Mission 3 — Surveillance des journaux et détection :

3.1 journald persistant et dimensionné :

Les journaux sont désormais conservés sur disque même après redémarrage, ce qui facilite les investigations ultérieures.

```
[root@srv-core ~]# systemctl restart systemd-journald
[root@srv-core ~]# journalctl --flush
[root@srv-core ~]# ls -ld /var/log/journal/$(cat /etc/machine-id)
drwxr-xr-x. 2 root root 28  4 nov.  16:08 /var/log/journal/22040c39fe4646149c3b957e19547e5d
[root@srv-core ~]# journalctl --disk-usage
Archived and active journals take up 8.0M in the file system.
[root@srv-core ~]# journalctl --verify | tail -5
PASS: /var/log/journal/22040c39fe4646149c3b957e19547e5d/system.journal
[root@srv-core ~]#
```

3.2 Requêtes d'investigation :

```
root@srv-core:~
[root@srv-core ~]# journalctl --list-boots
IDX BOOT ID                          FIRST ENTRY                 LAST ENTRY
 -1 4f7da6f846e14bf9bb3c1676257fe3a4 Wed 2026-11-04 12:19:02 CET Wed 2026-11-04 16:10:09 CET
  0 539cc8f1e90c4d1fbeab27b454f3746d Wed 2026-11-04 16:10:28 CET Wed 2026-11-04 16:15:02 CET
[root@srv-core ~]#
```

Last -F -n 10 : Affiche les 10 dernières connexions réussies.

```
[root@srv-core ~]# last -F -n 10
root     pts/0        192.168.56.1     Wed Nov  4 16:11:49 2026   still logged in
root     tty1                          Wed Nov  4 16:11:36 2026   still logged in
reboot   system boot  5.14.0-687.61.1. Wed Nov  4 16:10:32 2026   still running
root     pts/0        192.168.56.1     Wed Nov  4 16:06:34 2026 - Wed Nov  4 16:10:04 2026
(00:03)
root     pts/0        192.168.56.1     Wed Nov  4 14:03:12 2026 - Wed Nov  4 15:42:47 2026
(01:39)
root     pts/0        192.168.56.1     Wed Nov  4 12:23:06 2026 - Wed Nov  4 14:02:22 2026
(01:39)
root     tty1                          Wed Nov  4 12:19:35 2026 - down
(03:50)
reboot   system boot  5.14.0-687.61.1. Wed Nov  4 12:19:06 2026 - Wed Nov  4 16:10:08 2026
(03:51)
root     tty1                          Wed Nov  4 12:10:27 2026 - crash
(00:08)
reboot   system boot  5.14.0-687.61.1. Wed Nov  4 12:09:44 2026 - Wed Nov  4 16:10:08 2026
(04:00)

wtmp commence Mon Nov  2 12:01:18 2026
[root@srv-core ~]#
```

Lastb -n 15 : Affiche les 15 dernières tentatives de connexion échouées.

```
[root@srv-core ~]# lastb -n 15
clear    tty1                          Wed Nov  4 16:11 - 16:11  (00:00)
root     tty1                          Wed Nov  4 12:19 - 12:19  (00:00)
root     tty1                          Wed Nov  4 12:01 - 12:01  (00:00)
root     tty1                          Wed Nov  4 11:57 - 11:57  (00:00)
root     tty1                          Wed Nov  4 11:57 - 11:57  (00:00)
root     tty1                          Wed Nov  4 11:56 - 11:56  (00:00)
root     tty1                          Wed Nov  4 11:56 - 11:56  (00:00)
nadia                                  Tue Nov  3 11:42 - 11:42  (00:00)
nadia                                  Tue Nov  3 11:42 - 11:42  (00:00)
nadia                                  Tue Nov  3 11:42 - 11:42  (00:00)
nadia                                  Tue Nov  3 11:42 - 11:42  (00:00)
nadia                                  Tue Nov  3 11:42 - 11:42  (00:00)
nadia                                  Tue Nov  3 11:42 - 11:42  (00:00)
nadia                                  Tue Nov  3 11:40 - 11:40  (00:00)
nadia                                  Tue Nov  3 11:40 - 11:40  (00:00)

btmp commence Mon Nov  2 12:13:31 2026
[root@srv-core ~]#
```

journalctl -b -1 -p err --no-pager : Affiche les erreurs du démarrage précédent.

```
root@srv-core:~
[root@srv-core ~]# journalctl -b -1 -p err --no-pager
nov. 04 12:19:02 srv-core kernel: Warning: Unmaintained driver is detected: e1000
nov. 04 12:19:02 srv-core kernel: Warning: Unmaintained driver is detected: e1000_init_modul
e
nov. 04 12:19:02 srv-core kernel: Warning: Unmaintained driver is detected: e1000
nov. 04 12:19:02 srv-core kernel: Warning: Unmaintained driver is detected: e1000
nov. 04 12:19:02 srv-core kernel: Warning: Unmaintained driver is detected: e1000
nov. 04 12:19:08 srv-core kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be runn
ing on an unsupported hypervisor.
nov. 04 12:19:08 srv-core kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is l
ikely broken.
nov. 04 12:19:08 srv-core kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supp
orted graphics device to avoid problems.
nov. 04 14:02:16 srv-core systemd[1]: systemd-logind.service: Watchdog timeout (limit 3min)!
nov. 04 14:02:16 srv-core systemd[1]: systemd-udevd.service: Watchdog timeout (limit 3min)!
nov. 04 14:02:21 srv-core systemd-coredump[4333]: [↗] Process 647 (systemd-udevd) of user 0
dumped core.

                                                  Stack trace of thread 647:
                                                  #0  0x00007fe65d50fd0a epoll_wait (libc.so
.6 + 0x10fd0a)
                                                  #1  0x00007fe65da890d8 sd_event_wait (libs
ystemd-shared-252.so + 0x2890d8)
                                                  #2  0x00007fe65da8b3db sd_event_run (libsy
```

journalctl --facility=authpriv --since today --no-pager | tail -20 : **Affiche les moments d'authentication.**

journalctl _SYSTEMD_UNIT=sshd.service -g 'Invalid user|Failed|Accepted' --no-pager | tail -20 : **Montre les connexions SSH acceptées et refusées.**

3.3 Instrumenter auditd :

Les règles Auditd surveillent les événements sensibles du système. Les rapports générés permettent d'identifier rapidement les actions importantes.

```
[root@srv-core ~]# auditctl -l
-w /etc/passwd -p wa -k identity
-w /etc/shadow -p wa -k identity
-w /etc/group -p wa -k identity
-w /etc/gshadow -p wa -k identity
-w /etc/sudoers -p wa -k sudoers
-w /etc/sudoers.d -p wa -k sudoers
-w /etc/ssh/sshd_config -p wa -k sshd-config
-w /etc/ssh/sshd_config.d -p wa -k sshd-config
-w /usr/local/bin/helios-lastlog -p x -k helios-setgid
-a always,exit -F arch=b64 -S execve -F euid=0 -F auid>=1000 -F auid!=-1 -F key=privesc
-a always,exit -F arch=b32 -S execve -F euid=0 -F auid>=1000 -F auid!=-1 -F key=privesc
-w /etc/firewalld -p wa -k firewall
[root@srv-core ~]#
```

```
root@srv-core:~
[root@srv-core ~]# auditctl -s | head -5
enabled 1
failure 1
pid 751
rate_limit 0
backlog_limit 8192
[root@srv-core ~]#
```

ausearch -k privesc -i --start today | tail -20 :

```
root@srv-core:~
[root@srv-core ~]# ausearch -k privesc -i --start today | tail -20
----
type=PROCTITLE msg=audit(04/11/2026 16:24:37.379:141) : proctitle=/sbin/auditctl -R /etc/audit/audit.rules
type=SOCKADDR msg=audit(04/11/2026 16:24:37.379:141) : saddr={ saddr_fam=netlink nlnk-fam=16 nlnk-pid=0 }
type=SYSCALL msg=audit(04/11/2026 16:24:37.379:141) : arch=x86_64 syscall=sendto success=yes exit=1064 a0=0x3 a1=0x7ffeb3d3f250 a2=0x428 a3=0x0 items=0 ppid=2803 pid=2823 auid=roo
t uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=pts0 ses=3 comm=auditctl exe=/usr/sbin/auditctl subj=unconfined_u:unconfined_r:unconfined_t:s
0-s0:c0.c1023 key=(null)
type=CONFIG_CHANGE msg=audit(04/11/2026 16:24:37.379:141) : auid=root ses=3 subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 op=add_rule key=privesc list=exit res=yes
----
type=PROCTITLE msg=audit(04/11/2026 16:24:37.380:142) : proctitle=/sbin/auditctl -R /etc/audit/audit.rules
type=SOCKADDR msg=audit(04/11/2026 16:24:37.380:142) : saddr={ saddr_fam=netlink nlnk-fam=16 nlnk-pid=0 }
type=SYSCALL msg=audit(04/11/2026 16:24:37.380:142) : arch=x86_64 syscall=sendto success=yes exit=1064 a0=0x3 a1=0x7ffeb3d3f250 a2=0x428 a3=0x0 items=0 ppid=2803 pid=2823 auid=roo
t uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=pts0 ses=3 comm=auditctl exe=/usr/sbin/auditctl subj=unconfined_u:unconfined_r:unconfined_t:s
0-s0:c0.c1023 key=(null)
type=CONFIG_CHANGE msg=audit(04/11/2026 16:24:37.380:142) : auid=root ses=3 subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 op=add_rule key=privesc list=exit res=yes
[root@srv-core ~]#
```

aureport -k --summary -i :

```
root@srv-core:~
[root@srv-core ~]# aureport -k --summary -i

Key Summary Report
============================
total  key
============================
4  identity
4  sudoers
2  sshd-config
2  helios-setgid
2  privesc
1  firewall
[root@srv-core ~]#
```

3.4 Automatiser la détection :

```
[root@srv-core ~]# systemctl enable --now helios-watch-auth.timer
Created symlink /etc/systemd/system/timers.target.wants/helios-watch-auth.timer → /etc/sy
stemd/system/helios-watch-auth.timer.
[root@srv-core ~]# systemctl list-timers helios-watch-auth.timer
NEXT                         LEFT     LAST PASSED UNIT                    ACTIVATES      >
Wed 2026-11-04 16:36:00 CET 22s left -    -      helios-watch-auth.timer helios-watch-au>

1 timers listed.
Pass --all to see loaded but inactive timers, too.
lines 1-5/5 (END)
```

Script helios-watch-auth :

Le script analyse automatiquement les échecs d'authentification et applique des mesures de protection en cas d'activité suspecte.

```
[root@srv-core ~]# cat > /usr/local/sbin/helios-watchauth.sh <<'EOF'
#!/bin/bash
set -euo pipefail

LOCK=/run/helios-watchauth.lock
STATE=/var/lib/helios/blocked.txt
WHITELIST="192.168.56.1"

mkdir -p /var/lib/helios
touch "$STATE"

exec 9>"$LOCK"

if ! flock -n 9; then
    logger -t helios-watch-auth "Une instance est deja en cours"
    exit 2
fi

journalctl --since "-10 min" -u sshd --no-pager \
| grep "Failed password\|Invalid user" \
| awk '{print $(NF-3)}' \
| sort | uniq -c \
| while read -r count ip
do
EOFt 0  fi    "Blocage automatique de $ip ($count echecs SSH)"
[root@srv-core ~]# chmod 755 /usr/local/sbin/helios-watchauth.sh
```

Timer :

```
[root@srv-core ~]# cat > /etc/systemd/system/helios-watch-auth.timer <<'EOF'
[Unit]
Description=Lancement periodique helios-watch-auth

[Timer]
OnCalendar=*:0/2
Persistent=no
AccuracySec=10s

[Install]
WantedBy=timers.target
EOF
[root@srv-core ~]# systemctl daemon-reload
[root@srv-core ~]# systemctl enable --now helios-watch-auth.timer
```

```
[root@srv-core ~]# firewall-cmd --ipset=helios-blacklist --get-entries
192.168.56.11
[root@srv-core ~]#
```